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DEFT-Net: Explainable Deepfake Text Detection for Combating Information Disorder in the Age of Generative AI

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Abstract

The rapid evolution of Generative AI and large language models has drastically increased the risk of information disorder across digital communication platforms. The synthetically created messages with a deepfake aiming to mislead, manipulate the opinion of the masses, or disinformation is a serious danger to the honesty and credibility of online communication. Addressing this growing challenge requires an effective and transparent detection framework that can operate effectively at scale. This study introduces DEFT-Net (Deepfake Explainable FastText Tri-Convolutional Network), a practical and explainable deep learning architecture specifically designed for deepfake text detection on social media platforms. DEFT-Net leverages a multi-scale convolutional neural network (CNN) architecture combined with FastText embeddings to capture both syntactic and semantic patterns characteristic of synthetic content. The model was trained and evaluated using the *TweepFake* dataset, a rich collection of real and deepfake tweets, enabling the development of an efficient classifier capable of distinguishing authentic from fabricated text. To ensure transparency and foster trust in AI-based decision-making, this work integrates Explainable AI (XAI) techniques to provide interpretability of the model's predictions. Experimental results demonstrate that DEFT-Net achieves high detection performance attaining 93% accuracy, 92% precision, 95% recall, and a 93% F1-score, outperforming several evaluated baseline models on the TweepFake dataset using FastText, subword embeddings, BERT embeddings, and GloVe. The proposed framework can further aid in the overall combating of information disorder by providing a practical, explainable, and computationally efficient solution for deepfake text detection, which may have potential uses in digital forensics, online safety, and safeguarding information integrity.

Keywords Deepfake text detection · Information disorder · Generative AI · Explainable AI (XAI) · Convolutional neural networks (CNN) · Misinformation mitigation · NLP for disinformation · Synthetic content detection · Social media forensics · Trustworthy AI systems



1 Introduction

Social networking sites were created to enable individuals to interact and exchange their opinions and ideas through texts, images, audio, and videos [1]. Social media enables users to post content and comment on others' posts in real-time, without any preliminary limitation from the company offering the platforms or government agencies [2]. Social media posts and comments contain the views and perceptions of the users, and that attracts various stakeholders. People rely more and more on feedback received on social networks when making routine decisions, for example, when selecting a product to buy, a service to travel, an affordable hotel, or estimating the support level of specific political parties. Moreover, activities like customer support, competitor tracking, reputation management, and advertising are also used in businesses and organizational settings. Some of the widely used social media applications available online are: Twitter, Facebook, WhatsApp, YouTube, Instagram, Skype, LinkedIn, Viber, and WeChat. However, many researchers choose Twitter for their research since it is easy to use and has friendly policies for researchers [3]. For instance, Twitter users did not have to be friends to post, follow, or share content with individuals on the social platform. It allows users to send text messages that must not exceed 140 characters, known as short messages, with no restrictions. Some scholars consider this character limit as a disadvantage since it limits description, but others could find this as an advantage as they like fast interactions.

Software programs called bots execute operations on artificial intelligence (AI)-based social media accounts by accurately simulating user actions that involve content interaction activities while using text generation and find-replace methods and deepfake production techniques [4]. Autonomous feature extraction and learning from raw data is a specialization of machine learning that falls under the domain of deep learning. The conceptual fusion of deep learning and fake technology gives rise to deepfake which describes AI-generated deceptive multimedia content ranging from text to images and audio and video formats [5]. Deepfake text describes AI-generated or altered linguistic content that imitates human communication while intending to create authentic communication. Deepfake content distribution on social media platforms causes widespread concern in multiple domains because it fools users with fake representations of human-generated content [6]. It highlights the urgent need for reliable and intelligent detection mechanisms to preserve information integrity and user trust online.

Generative pre-trained transformers (GPT) are one of the categories of the large language models developed based on the transformer architecture. Furthermore, the changes reflected in the architecture of the GPT-n series of models created by OpenAI are also discussed. Prior and later generations of GPT relied on the successes of the transformer architecture, as the self-attention mechanism outperformed earlier RNN-type models. The generative pre-training as the third approach to model training involves learning language patterns by preparing the text corpus for pre-training activities. The model can then be fine-tuned for the task in hand, such as translation, using a small labelled dataset [7]. GPT-2 [8], and GPT-3 are two of the leading NLP-based models that were introduced by OpenAI. The AI-powered ChatGPT system that OpenAI designed uses GPT-3.5 along with GPT-4 computational architectures and serves as an evolution of the InstructGPT model [9]. The launch of this technology occurred in late 2022 as it immediately drew worldwide recognition and substantial interest.

Social media platforms mainly produce text-based deepfakes through GPT models because the models provide better short-text generation capabilities than GROVER [10] and CTRL [11]. The detection of content made by GPT-2 proves significantly challenging relative to content generated from RNNs or prior methods [12]. The study dedicated efforts to analyzing deepfakes produced by RNNs in combination with GPT-2 and additional robot software technologies as part of its research strategy. The study investigates the development of advanced detection techniques suitable for recognizing deepfake text among time-sensitive social media content. The proliferation of deepfake text on social media platforms presents a significant challenge due to its ability to mimic human writing styles convincingly. These AI-generated texts are often used maliciously to mislead, manipulate, or deceive users and threaten online users, so detection has become a top priority. Twitter's brief content style makes detecting created texts harder for social media platforms. The primary objective of this research is to analyze and address the challenges in identifying deepfake content on social media sites. The brief and informal character of social media

material and the usage of slang, acronyms, and uncommon terms, as well as the human communication styles of generative AI models, are some of these difficulties. The paper suggests creating a deep learning framework that uses effective word embedding methods suited for short social media text to capture semantic links better. Furthermore, the study incorporates interpretable AI techniques within the framework, guaranteeing that the model's decision-making process is transparent. This method increases confidence in the system's capabilities and clarifies the model's decision-making process. While traditional approaches to fake or manipulated text detection often rely on standard CNNs or transformer-based architectures, this study introduces DEFT-Net (Deepfake Explainable FastText Tri-Convolutional Network)—an engineering-driven and application-oriented explainable framework specifically designed for detecting deepfake textual content on social media. DEFT-Net is unique in three aspects. First, it uses a multi-scale CNN architecture, which achieves a variety of semantic and syntactic regularities at multiple convolutional scales, making it more sensitive to fine manipulations of short, noisy tweets. Second, it incorporates FastText embeddings to make use of information on the subword level to achieve better generalization to informal and morphologically diverse language, as is common to Twitter data. Third, DEFT-Net uses the interpretability with SHAP, which allows getting a clear explanation of why a model made this choice—a feature that is rarely discussed in prior studies on deepfake text detection. Together, these design choices bridge a key research gap by delivering an interpretable, lightweight, and domain-optimized model that enhances both accuracy and explainability compared to existing deep learning frameworks. The direction of this study was determined by the following research questions.

- RQ1: How can deepfake text, particularly short-form social media messages, be effectively distinguished from authentic human-written content?
- RQ2: Which word embedding techniques (FastText, FastText Subword, BERT, GloVe) provide the most effective representation for deepfake text detection?
- RQ3: Which deep learning architectures (CNN, LSTM, CNN-LSTM, BiLSTM+Attention, RCNN) demonstrate superior performance in detecting deepfake text across noisy, short social media content?
- RQ4: How can explainable AI (XAI) techniques, specifically SHAP, enhance transparency and interpretability in deepfake text detection models?

This study contributes a practical deepfake text detection framework with the following application-oriented contributions:

- Developed methods to capture complex semantic relationships in text data using advanced word embeddings, enhancing the understanding of the language used in deepfake text.
- Proposed a DEFT-Net (Deepfake Explainable FastText Tri-Convolutional Network) framework combining Convolutional Neural Networks (CNNs) and word embeddings, offering improved interpretability and practical generalization within the evaluated social media setting for detecting AI-generated text.
- Integrated XAI techniques to enhance the transparency and trustworthiness of the detection model, providing insights into the decision-making process and enabling further research.
- Conducted a comprehensive evaluation of machine learning and deep learning models, establishing a performance benchmark for deepfake text detection, particularly in the context of short social media content.

The remaining structure of the thesis is as follows: Sect. 2 offers an extensive examination of past research efforts concerning the detection of deepfake text. Section 3 introduces and discusses the techniques and methodologies employed in this investigation. It progresses by presenting the proposed methodology, along with a concise explanation of the machine learning and deep learning models. Section 5 explores the outcomes of the experiments and presents a detailed comparison between the cutting-edge learning classifiers and the proposed approach. Section 6 concludes with a summary of the results generated by machine learning and deep learning models in Conclusion and Future Work.

2 Related Work

Deepfake technologies first appeared and developed in the field of computer vision [13–15], but have been gradually evolving in fully-fledged attempts at audio manipulation [16, 17] and text synthesis [18]. In the computer vision domain, deepfakes refer to subject manipulation, particularly of the face: whole-face synthesis, identity exchange, attribute alteration, emotion switching [19], and body reenactment [20]. Relatively new audio deepfakes generate spoken speech from text by mimicking other people’s voices after listening to them for only a short time [16].

The succeeding transformer-based language models are GPT [7], BERT [21], GPT-2 [18], GPT-4 [22], LLaMA [23], etc.) improved not only language-generating tasks but also natural language interpretation tasks. With each new improvement in coherent text generation, the generated texts become more like human writing. These models use complex architectures that include self-attention techniques and large-scale pre-training on massive datasets. As a result, they have shown amazing capabilities in a variety of areas, including conversational agents, machine translation, and sentiment analysis. With each iteration, these models increase the generated text’s fluency, coherence, and contextual knowledge, making it increasingly similar to human writing.

The most popular deepfake text detectors may be divided into two main categories: the need for statistical methods and optimization of existing models. These two groups encompass a wide spectrum of techniques whereby each has unique own strengths and limitations. It is possible to enhance a model’s ability to differentiate the text created by a machine and a human-written text by fine-tuning the existing models on specified datasets or tasks linked to deepfake text detection. These models, like BERT and RobertA as pre-trained models, are transformer [24]. On the other hand, statistical models employ statistical analysis and pattern recognition to identify unique features or anomalies indicative of machine-generated material such as DetectGPT [25], Giant Language Model Test Room (GLTR) [26], and MFD [27] enable the automatic detection of peculiar parts or deviations from the ordinary in AI-generated content. Both solutions have been analyzed and optimized to fit the growing needs that arise with the increasing usage of AI content in several fields.

Besides watermarking techniques, decision trees, XGBoost, as well as transformers have also been utilized to differentiate between text written by humans and AI. In their work [28], the authors examined stylometric features of texts synthesized by GPT–3.5 and the recently disclosed GPT-4 and contrasted them with the texts initially authored by people. This paper analysed Japanese writings and applied multi-dimensional scaling (MDS) to compare the stylistic differences between generated AI and human writings. This study examined Japanese writings and utilized multi-dimensional scaling (MDS) to map the stylistic distinctions between AI-generated and human-written content. The researchers examined numerous significant stylometric variables, including bigrams of parts of speech, bigrams of postpositional particle words, comma placement, and function word rate. These attributes indicated unique distributions across GPT-generated and human-authored texts, suggesting that, despite GPT-4’s broader parameter range, AI-generated texts may be distinguished from human writing based on stylometric patterns. This research also focused on Random Forest (RF) classifier algorithms to distinguish between AI and human writings by evaluating stylometric features. Specifically, the RF classifier trained on the number of times function words only achieved an accuracy of 98%, and all the stylometric factors employed in the classifiers achieved an accuracy of 100%. Stylometrics analysis can properly separate content generated by AI from content written by humans in Japanese text. The study showed that RF detected AI content, which supports fact-checking teams to fight misinformation across different languages.

Another novel approach has used intrinsic dimension analysis techniques, such as the Persistent Homology Dimension Estimator (PHDE), to successfully distinguish between human-generated and AI-generated information. The authors [29] investigated the problem of distinguishing AI-generated text from human-authored material due to its increasing complexity. The research team introduced a concept known as intrinsic dimensionality that remains stable across all text forms when written by experts of different levels. This method aims to determine the basic dimensional qualities of the geometric patterns produced from embedding points that represent

text samples. Text in alphabet systems requires an intrinsic dimensionality of 9 but Chinese text needs only 7. AI systems create text output that needs about 50% fewer dimensions than human-created text.

For the detection of AI-generated tweets, Gambini et al. [30] proposed a three-branch network that integrates both syntax and semantics into a model that improves over the fine-tuned BERTweet. Progress in this way is particularly significant in view of the enormous breakthroughs in natural language generation, which, in turn, lead to the creation of professional language generators that synthesize text that is rather similar or even identical to the text written by a human. As generative capabilities advanced, concerns have been raised over the utilization of these capacities, which pose threats such as the spread of fake news, copying materials, and learning interruption in particular settings.

SeqXGPT [31] presents a novel methodology for identifying AI-generated material by the use of sentence-level log probability metrics obtained from Large Language Models (LLMs). The enhanced capability of LLMs to generate human-like language has intensified apprehensions about their potential for misuse, especially in the creation of misleading or fraudulent information. According to SeqXGPT researchers, effective detection systems need to analyze sentences to reveal AI-generated text.

DetectLLM [32] offers important observations regarding the relationship between text produced by AI and the foundational structure of LLMs. The researchers discovered that the AI text most likely appears at regions of the log probability function of language models where the graph is turning down. FastDetect-GPT further improved the existing methods by maximizing the text detection procedure of AI-generated text, therefore, eliminating the need to perform perturbation analysis. Rather, it underscored the utilization of conditional probability curves, facilitating a more effective assessment of text authenticity. This novel method highlighted a significant difference: writings generated by AI generally have elevated conditional probability curves in comparison to those written by humans.

The researchers in [33] developed a zero-shot detection technique using GECScore to separate human-written content from material produced by LLM systems. Their detection approach shows better results than the top zero-shot and supervised detectors across various adversarial conditions. DetectGPT by [25] employs logarithmic analysis to identify deepfake news articles more effectively than other zero-shot detectors in this field.

To explore text classification in depth, the researchers have applied explainable AI with machine learning models. The authors [34] presented an Explainable Text Classification Model designed for identifying COVID-19-related misinformation by combining the Long Short-Term Memory (BiLSTM) model with Local Interpretable Model-agnostic Explanations (LIME). While attaining high classification probabilities, both schemes were beneficial in that BiLSTM achieved a higher classification accuracy than LSTM, as well as LIME, enhancing the interpretability of the model, which is a disadvantage with neural networks, where models act as a black box. We used an integrated model of LIME and BiLSTM to achieve these goals, as it helped in providing an insight into the prediction process, which is a vital component while deploying such models for false news detection, where model trust is of paramount importance.

In another study, [35] examined the extent to which model explainability can help in the classification of human written and LLMs composed documents. The study reviewed the conventional two way classification strategy and proposed a three way classification system, that is, including an 'indecisive' class for texts which could be classified into either source. This distinct method enhances the features of findings concerning detection which makes them easier for a common user to understand. Through the comparison of various datasets and the best detectors, the authors highlight the necessity of using comprehensive descriptions along with categorization, enabling the recognition of critical factors in improving the explainability of text detection systems created with AI technologies.

The authors [36] investigated the problem of justifying predictions produced by machine-generated text detectors, addressing the increasing difficulties in distinguishing between content created by AI and that written by humans. They underlined that, in many circumstances, simply delivering a black-box prediction is insufficient, and it is necessary to explain the reasoning behind the detector's choice. To this end, they evaluated explanation techniques that measure feature importance, centred on how specific components of a text affect a detector.

Classical and contextual word embeddings remain strong baselines for misinformation detection. A comprehensive IEEE Access benchmark compared CNN/RNN/RCNN-style architectures across multiple corpora while varying embeddings (Word2Vec, FastText, GloVe), showing that representation choice strongly mediates downstream accuracy and robustness in news veracity tasks [37]. Follow-up studies further contrasted static vs. context-aware embeddings and highlighted subword-aware representations (e.g., FastText) as particularly effective when text is noisy or short, as on social media [38]. Domain-specific investigations (including Arabic and multilingual settings) also report gains when pairing pre-trained embeddings with lightweight classifiers for fake-news detection in the wild [39].

Transformer encoders are widely adopted in fake-news pipelines. MisRoBERTa ensembles BART and RoBERTa to discriminate real vs. multiple fake categories, outperforming single backbones on diverse benchmarks [40]. Independent evaluations comparing BERT/ALBERT/RoBERTa corroborate that robust pre-training and deeper attention stacks typically yield higher F1—albeit with greater latency and memory costs, which can constrain real-time deployment at scale [41].

Beyond token-level encoding, higher-level embeddings (e.g., document encoders and sentence transformers) often provide stronger separability for news-article classification. A recent study using document embeddings reported consistent gains over more complex deep networks across five large news corpora, arguing that document-level encoding is a dominant factor for accuracy in fake-news classification. [42] These results motivate our exploration of representation choice (FastText vs. BERT/GloVe) within DEFT-Net.

Ensemble and expert-based architectures increasingly fuse heterogeneous signals. DANES introduces a dual-branch ensemble—Text branch (content) + Social branch (propagation/user context)—and learns a joint network embedding to improve classification, demonstrating the value of social signals alongside text [43]. GETAE (Graph Information Enhanced Deep Neural Network Ensemble) extends this idea with a Text Content Embedding (lexical + syntactic features) and a Propagation Embedding that captures diffusion patterns; combining graph/node embeddings with text features further boosts detection and early-stage identification [44]. Beyond these, the misinformation community has explored mixture-of-experts style and task-ensemble approaches across recent shared-task venues (e.g., CLEF/PAN working notes), reflecting a broader shift toward modular architectures that specialize by modality, domain, or signal type.

Positioning detection within a mitigation ecosystem, network immunization strategies proactively curb spread by strategically “inoculating” or throttling high-influence nodes. Community-based algorithms (e.g., CONTAIN) select users at community boundaries and origin clusters to minimize cross-community diffusion, outperforming degree- or betweenness-only heuristics on social graphs [45]. In parallel, real-time pipelines aim to surface harmful content early. Cloud/streaming systems (e.g., FANDC) and online-learning approaches have been proposed to continuously classify high-velocity social streams, addressing limits of static, batch-trained detectors and improving time-to-mitigation in production [46]. Recent surveys also emphasize multimodal cues (text + image/video + network) and recommend joint detection-mitigation loops (prebunking, warning prompts, targeted outreach) to translate detection signals into behaviorally meaningful interventions [47].

There are several constraints and challenges in the detection of textual deepfakes. One of the major issues is adversarial generation where the generative models adapt to the detection techniques resulting in a constant war between detection and generation strategies. Moreover, data imbalance is another problem that hinders the process, since there are few labelled datasets with multiple examples of deepfakes in text form to train the detection models. Another is the problem of semantic stability; the generative models are really very powerful that the output generated by them can be semantically correct and semantically significant in context and therefore makes it even harder for artificial intelligence to differentiate between real and fake information.

Thus, the proposed work aims at filling these gaps by integrating feature engineering: those features of stylometry and structural to improve classification model evaluation. Moreover, this research aims to address the method of deepfake detection for performing better than the supervised detectors and to overcome their drawbacks in the dynamically changing AI landscape. Using contemporary techniques and data analysis, this research tries to develop objectives of deepfake text detection technology and that will be very helpful for the research fraternity.

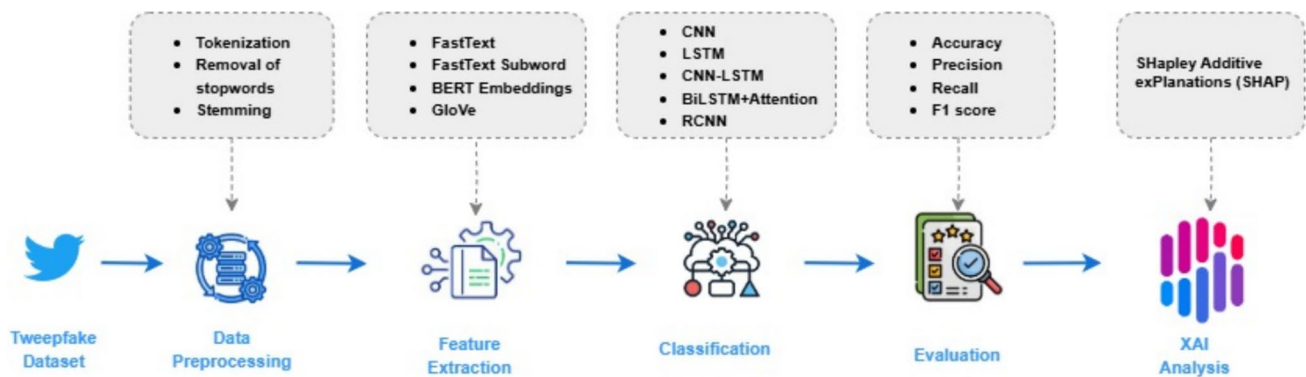
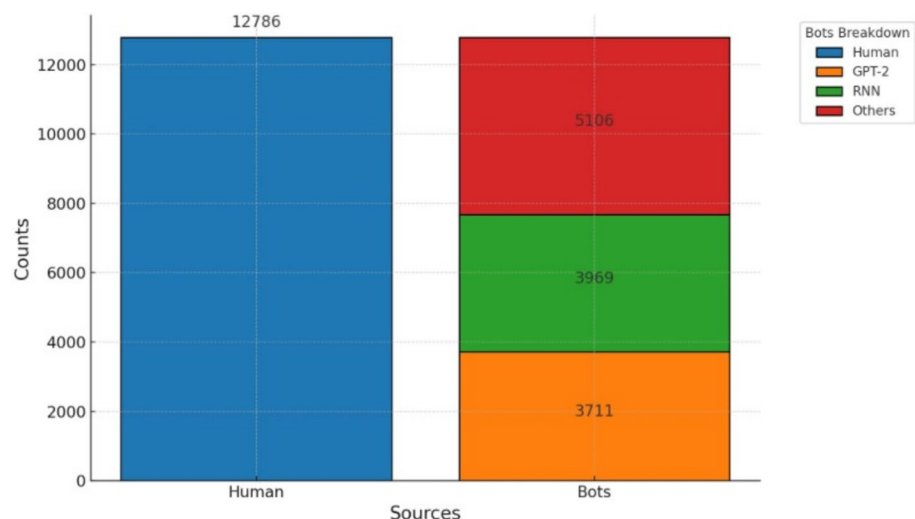


Fig. 1 Architecture of methodologies adopted for deepfake tweet classification

Fig. 2 Countplot showing account-type data distribution



3 Material and Methods

This section presents a thorough examination of the dataset utilized in this research. Additionally, it introduces and discusses the methods and techniques employed in the experiments. The main emphasis of this study is the evaluation of deepfake tweets. To initiate the experiments, tweets were gathered from Twitter and underwent preprocessing to eliminate extraneous information. Subsequently, features were extracted using techniques: TF, TF-IDF, FastText, and FastText subwords. Following this, tweet classification was conducted applying diverse machine learning and deep learning algorithms. The effectiveness of the proposed model was assessed using criteria like accuracy, precision, recall, and F1 score. Furthermore, SHapley Additive exPlanation (Shap) was leveraged to illustrate the findings of the proposed strategy. The experimental approach is illustrated in Fig. 1.

3.1 Dataset

This research employs the TweepFake dataset [48], which includes 25,572 tweets collected from 40 accounts 17 belonging to humans and 23 to bots. Each entry is labeled to indicate its origin, distinguishing between human-generated content (12,786 tweets from 17 accounts) and bot-generated text. The bot-produced tweets are further categorized by generation method: GPT-2 (3,711 tweets from 10 accounts), RNN (3,969 tweets from 5 accounts), and other techniques (5,106 tweets from 5 accounts). A count plot in Fig. 2 illustrates the account-type distribution of the data set.

3.2 Text Preprocessing

We thoroughly cleaned and prepared the data prior to providing the tweets to the embedding and classification models to ensure consistent data and eliminate noise as well as enhance the extraction of meaningful features. The preprocessing steps were designed according to widely adopted natural language processing (NLP) best practices for social media text cleaning and normalization [49]. The complete sequence is described below:

Lowercasing: All characters in the tweets were converted to lowercase to minimize case sensitivity and reduce redundant word variants (e.g., AI and ai treated equally).

Punctuation and Non-Alphanumeric Filtering: Non-informative punctuation marks, special characters, and digits were removed except where apostrophes convey linguistic meaning (e.g., “don’t”).

Tokenization: Tweets were tokenized using the NLTK tokenizer, which effectively handles informal constructs, contractions, and slang commonly found in microtexts.

Common English stopwords (e.g., “the,” “is,” “in”) were removed using the NLTK stopword list to focus on semantically meaningful terms. However, function words indicative of personal expression (e.g., “I,” “my,” “you”) were retained, as prior studies indicate their discriminative importance in human-authored text [50]

Lemmatization: Each token was reduced to its lemma form using WordNet lemmatizer to unify word inflections (e.g., running, ran → run), improving generalization without losing context.

After preprocessing, the cleaned and tokenized tweets were passed to the embedding layer (FastText, BERT, GloVe, etc.), ensuring that the input to the deep learning models captured meaningful and consistent linguistic features.

3.3 Feature Extraction

Feature extraction methods are necessary to train deep learning models. In this work, to transform textual data into numerical features, three embeddings were employed: FastText, GloVe, and BERT. These embeddings were selected to capture lexical, semantic, and contextual information from tweets.

The selection of these embeddings is supported by both theoretical reasoning and empirical validation from prior studies on short, informal, and noisy social media texts. In particular, FastText was demonstrated to be more effective at sentiment analysis and detection of fake news on social platforms and where misspellings and non-standard forms of words are prevalent [51, 52]. GloVe on the other hand has shown consistent and steady performance on general text comprehension and classification tasks on twitter data sets [53]. In the meantime, BERT has been found to be very effective in the context in capturing contextual nuances and in disambiguating linguistic ambiguity and sarcasm in short text, being more effective than traditional embedding models [54]. In general, the strategy adopted in the suitability of embedding meets the past literature on the topic, where the combination of several embedding techniques in the textual analysis in social media improves the accuracy and the validity of the classification results [55, 56].

3.3.1 FastText

FastText is a publicly available toolkit created by Facebook AI Research (FAIR) for learning word embeddings and executing classifications. It features 600 billion word vectors and 2 million frequently used words, each represented in 300 dimensions. Besides single words, it incorporates manually engineered n -grams as features. Due to its simple architecture, text classification can be performed with high efficiency and accuracy [57]. This method supports the development of both supervised and unsupervised learning models that generate word embeddings. It can quickly process large-scale datasets and perform well in tasks such as semantic similarity analysis and text categorization. Various word embedding strategies have been used to address different text categorization difficulties. Pre-trained word embeddings anticipate word contexts in an unsupervised way, assuming

that surrounding words have similar meanings. Among these strategies, FastText stands out for using subword information to improve the representation of rare and complicated words via morphological analysis.

This capacity helps to improve its generalizability. FastText word embedding creates vectors from 'n-grams', allowing it to handle out-of-vocabulary words. In this study, pre-trained FastText embeddings with 300-dimensional vectors were used. The embeddings were trained on a vast corpus, providing rich semantic information for words and subwords. These embeddings were integrated into the classification tasks to transform textual input into dense vector representations, which were subsequently fed into the deep learning model for feature extraction and classification.

3.3.2 FastText Subword

The FastText Subword model generates word embeddings by analyzing subword components, trained on a large corpus of 600 billion tokens from Common Crawl, resulting in 2 million distinct word vectors [58]. Unlike traditional word-level embeddings, this approach breaks words into smaller units, enabling richer semantic representations. By incorporating subword information, FastText extracts the meaning of morphemes and affixes, and thus is most useful with morphologically complex languages. This enables the model to respond to changes in verb conjugations, pluralization and gender inflections better.

3.3.3 BERT Embeddings

To explore the impact of contextualized word representations on deepfake text detection, this study incorporated BERT (Bidirectional Encoder Representations from Transformers) embeddings as an embedding input layer in our deep learning pipeline [59]. The transformer-based model called BERT is capable of extracting deep bidirectional context of the words before and after words in a sentence which provides a great boost compared to the previous static word embeddings.

The pre-trained BERT base model (bert-base-uncased) was employed to generate dense vector representations of input social media messages. Each message was tokenized using BERT's tokenizer, which includes WordPiece tokenization and the addition of special classification tokens (e.g., [CLS], [SEP]). The resulting token sequences were then passed through the BERT model to extract contextual embeddings. For each input message, we extracted the embedding corresponding to the [CLS] token, which represents a holistic summary of the entire message. These [CLS] token embeddings (768-dimensional for BERT base) were used as fixed input features to the classifiers. This setup allowed the model to benefit from the rich semantic features captured by BERT while delegating the classification task to a lightweight, task-specific architecture.

3.3.4 GloVe

GloVe (Global Vectors for Word Representation) is an unsupervised learning algorithm that constructs word embeddings by aggregating global word-word co-occurrence statistics from a large corpus. Unlike contextualized models, GloVe produces a single vector representation for each word, regardless of its context. A study also used GloVe embeddings [60] to evaluate the impact of pre-trained static word representations on the detection of deepfake messages. For this experiment, the pre-trained GloVe 6B embeddings with 300-dimensional vectors are utilized, which were trained on a corpus consisting of 6 billion tokens from Wikipedia and Gigaword. Each word in the input social media messages was converted into its corresponding GloVe vector. Words that were not found in the GloVe vocabulary were initialized with random vectors following a uniform distribution.

3.4 Deep Learning Models

The experimental framework incorporates several deep learning architectures, specifically Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), a hybrid CNN-LSTM model, BiLSTM+Attention, and BERT-base.

3.4.1 CNN

The complete CNN framework typically incorporates several specialized layers: convolutional layers for feature extraction, pooling layers (employing either max or average operations) for dimensionality reduction, fully-connected layers for final classification, and dropout layers for regularization. The network processes information in a feed-forward manner, with each layer's output serving as input to subsequent layers. As shown in Eq. 1, the Rectified Linear Unit (ReLU) activation function is commonly implemented to introduce non-linearity into the model.

$$y = \max(0, i) \quad (1)$$

The activation function transforms input i into output y through nonlinear mapping. Feature extraction in convolutional layers is achieved through parameterized kernel weights that evolve during the training process. The objective function employs cross-entropy, with its complete mathematical formulation presented in Eq. 11.

$$\text{crossEntropy} = -(i \log(p) + (1 - i) \log(1 - p)) \quad (2)$$

Where p indicates the model's probability estimates and i corresponds to the ground truth labels. A sigmoid-based error function is implemented for CNN output generation, offering improved gradient flow compared to standard backpropagation. The network is configured for two-class classification tasks.

3.4.2 LSTM

LSTM is among the advanced deep learning techniques commonly used to tackle text classification problems. The LSTM architecture employs three specialized gating mechanisms: Input gate i_k controlling new information flow, Forget gate f_k determining which information to discard, and Output gate o_k regulating information exposure.

These gates collectively manage information retention through a memory cell c_k , preserving relevant patterns while eliminating noise according to the specified dropout parameters. Among various LSTM implementations, this study adopts the specific formulation presented in Eq. 3, 4 and 5.

$$i_k = \sigma(W_i s_k + V_i h_{k-1} + b_i) \quad (3)$$

$$f_k = \sigma(W_f s_k + V_f h_{k-1} + b_f) \quad (4)$$

$$o_k = \sigma(W_o s_k + V_o h_{k-1} + b_o) \quad (5)$$

$$c_k = \tanh(W_c x_k + V_c h_{k-1} + b_c) \quad (6)$$

The weight matrices W and V contain the trainable parameters of the model. The system state is characterized by: s_k representing the temporal input at step k . h denotes the hidden state through $k-1$ time steps, b presents the bias term; The memory cells c undergo updates at each time step k . The architecture implements full connectivity between the LSTM's final hidden layer and all neurons in the subsequent dense classification layer.

3.4.3 CNN-LSTM

In ensemble methods, the models tend to perform better as compared to individual models. Due to the benefits of integrating CNN's ability to do automatic feature extraction with LSTM's ability to learn the long-term temporal relationship, the CNN-LSTM duo has become prevalent. As a result, the LSTM layers are in a position to comprehend temporal relations because of the right function representations given out. CNN-LSTM is a successful solution for solving time series and classification issues [61].

3.4.4 BiLSTM+Attention

To effectively capture both sequential and contextual dependencies in social media messages, this study employed a Bidirectional Long Short-Term Memory (BiLSTM) network augmented with an Attention mechanism as the core deep learning architecture for classification [62]. The BiLSTM+Attention model was integrated with multiple word embedding techniques including BERT, GloVe, FastText, and FastText subword—to assess its performance under different representation schemes.

The BiLSTM layer also allows the model to read through the sequences of words in both directions, which allows the model to recognize long-range dependencies and contextual cues on both ends of a sentence and both forward and backward words. This is especially crucial when it comes to identifying minor patterns that can distinguish between deepfake messages and original messages.

Once the embedded input sequences (derived from the respective embedding techniques) were passed through the BiLSTM layers, the resulting hidden states were fed into an Attention layer. The Attention mechanism assigns learnable weights to different time steps in the sequence, allowing the model to focus on the most informative parts of the text when making a classification decision. This is especially valuable in social media texts, where specific phrases or tokens may carry disproportionate importance in identifying generated content.

The architecture of the BiLSTM+Attention model included:

Embedding Layer: Initialized with pre-trained embeddings (BERT, GloVe, etc.) depending on the experiment.

BiLSTM Layer: Captures bidirectional sequence dependencies using 128 hidden units in each direction.

Attention Layer: Computes context vectors using attention scores over BiLSTM outputs.

Dropout Layer: Applied with a rate of 0.5 to prevent overfitting.

Dense Layer: A fully connected layer with ReLU activation.

Output Layer: A softmax activation function for binary classification (real vs. deepfake).

Adam optimizer was utilized to train the model and the loss function was categorical cross-entropy with a learning rate of 0.001. The training was done over 20 epochs and early stopping was done using a validation loss in order to provide maximum generalization.

By combining BiLSTM with an attention mechanism, the model can focus on the most important parts of each message, which improves classification accuracy across different embedding methods. The overall performance of FastText embeddings applied to the BiLSTM-Attention architecture was highest in comparison with all the other tested setups, but BERT and FastText subword embeddings were also highly effective.

3.4.5 RCNN

To capture both local features and long-term dependencies in text, we employed a Recurrent Convolutional Neural Network (RCNN) [63]. This hybrid model combines the sequential modeling strength of a BiLSTM with the local feature extraction capability of 1D CNN.

The input text was first converted with pre-trained word embeddings (GloVe, FastText, FastText subword and BERT embeddings). A BiLSTM layer was used to extract the contextual relationship in both directions to the embeddings. The convolutional operations were followed by the max-pooling which minimized the dimensions and retained the most important features. The resulting feature map was flattened and run through a fully connected dense layer and then a softmax activation function to do binary classification (real vs. deepfake).

The model was trained using the Adam optimizer, with a categorical cross-entropy loss function, and early stopping was employed based on validation loss to prevent overfitting.

This RCNN architecture effectively combined the local feature extraction power of CNN with the sequential understanding capability of RNNs, resulting in a model that was both expressive and robust. Experimental results demonstrated that RCNN achieved competitive performance when combined with contextualized word embeddings, particularly in detecting subtle linguistic cues present in deepfake social media messages.

3.5 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) is a growing field in artificial intelligence that aims to make machine learning models more interpretable and transparent. The main aim of XAI is to make people comprehend what makes AI models make certain decisions in order to trust these findings. The requirement to comprehend how such models generate results has been of high importance as AI technology is advancing in the most important sectors such as healthcare and finance. In contrast to the more common type of black box models, which make decisions without explaining how the decision is made, XAI lets you see the inner workings of complex models such as deep neural networks.

XAI is necessary for a variety of reasons, including ethical considerations, legal requirements, and the practical need for accountability in AI-driven judgments. People need to trust AI technology before organizations will accept it for use in critical applications. People will trust AI when XAI shows them clear reasons for decisions while producing results that match human understanding. Explainability helps to uncover biases that might be present in the data or model. With XAI, stakeholders can understand whether the model is fair, and if certain groups are unfairly disadvantaged by the AI's decisions, corrective measures can be applied. By using XAI approaches, data scientists can locate their model's shortcomings and enhance their AI systems through better adjustments and enhancements. There are several techniques to achieve explainability in AI; this study employs SHAP (SHapley Additive exPlanations).

3.5.1 SHAP (Shapley Additive Explanations)

SHAP (SHapley Additive exPlanations) is one of the most effective and widely used interpretations of the predictions of machine learning models [64]. Cooperative game theory serves as its foundation, and there is a straightforward method for allocating each characteristic's contribution to the overall projection. In order to further advance knowledge of how AI choices are produced, SHAP aims to give both local and global interpretability and convert model outputs into clear, useful explanations. SHAP uses the Shapley value concept from cooperative game theory to determine a participant's contribution to total payoffs. In machine learning, the "players" are the input variables, while the "payoff" is the product of the given model's prediction. Accordingly, the Shapley value determines every potential combination of qualities that might be present in the prediction as well as their contributions depending on whether or not they are present. The SHAP methodology is defined by several fundamental properties:

- The sum of SHAP values for all characteristics is equal to the difference between the predicted value and the overall mean in the set.
- In the case of two characteristics having an equal impact on a prediction, both get the same SHAP values.
- If a characteristic does not have an impact on the outcome then its SHAP value is 0.
- The SHAP values of several features can be summed up to produce the overall prediction.

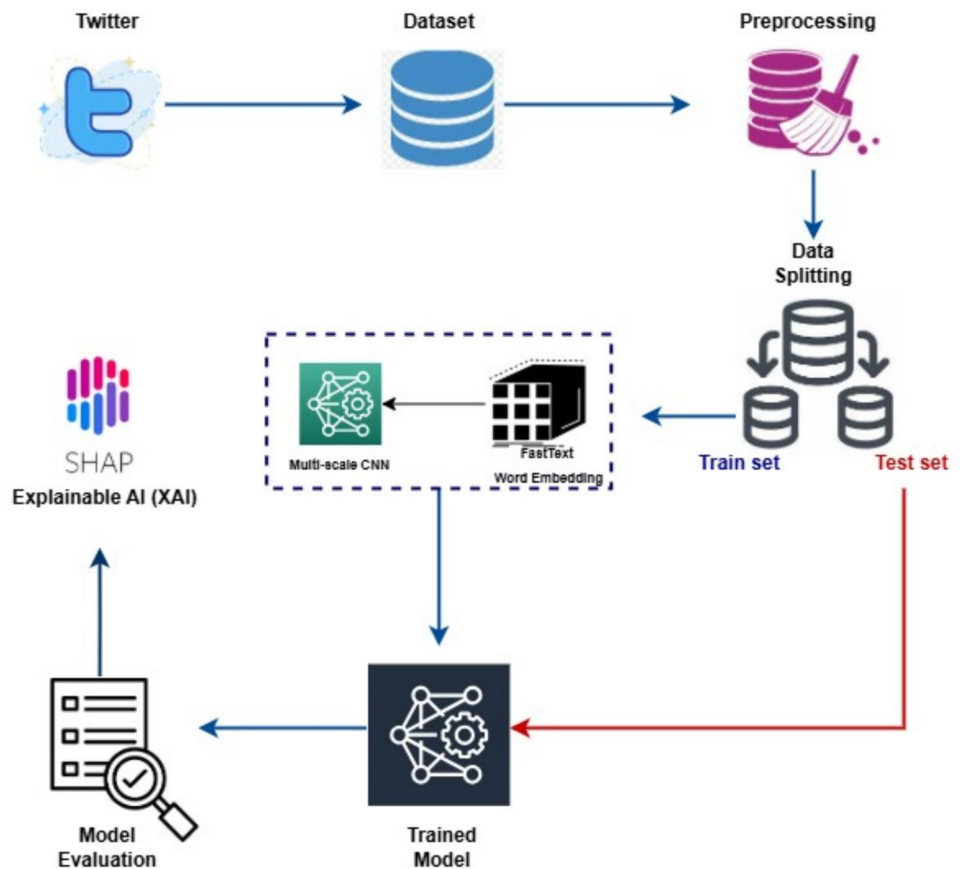
The Shapley value for a feature i is computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (7)$$

where N is the set of all features, S is a subset of N excluding features i . $f(S)$ is the model's prediction based on the features in subset S . $|S|!$ and $|N|$ denote factorials to account for permutations. This formula computes the marginal contribution of a feature i by considering all possible subsets S of features and averaging the contributions.

The SHAP workflow involves selecting a model, preparing and preprocessing data, training the model, calculating SHAP values for predictions, and interpreting the results using visual tools to understand the influence of each feature. SHAP performs very well in text identification and natural language processing (NLP). For text-based applications, what SHAP provides in terms of interpretability can enhance the understanding and trust towards model outputs. SHAP may be utilized for diverse text detection tasks like sentiment analysis, spam identification, and misinformation detection by explaining the keywords or phrases that contribute to model predictions. This enables users to gain insights into the decision-making process of the model, improving understanding and refinement in tasks like named entity identification and deepfake text detection.

Fig. 3 Architecture of proposed DEFT-Net for deepfake tweet classification



3.6 Proposed Methodology

This section describes the methods used to classify tweets by taking advantage of the potential of deep learning, especially a Convolutional Neural Network (CNN). Figure 3 illustrates the DEFT-Net architecture, which integrates multi-scale CNN with FastText word embeddings to improve tweet categorization accuracy. Furthermore, SHAP (SHapley Additive Explanations) is used to provide explainability in model judgments.

The proposed method starts with the collection of a labelled dataset from a public repository, containing tweets from both human and bot accounts. The training and the evaluation of the model performance are based on the dataset. As the model may only accept well-formatted text, the input data undergoes a number of preprocessing steps before being passed into the model. Cleaning the data also contributes to removing the noise that is not necessary and the CNN is able to concentrate more on meaningful data in the analysis process. The text is then cleaned and transformed into numeric form, which is also essential in providing data to the neural network. Embeddings of words are FastText word embeddings, which are utilized in this process to convert tokenized tweets into meaningful vectors. FastText is preferred for its ability to capture subword information and contextual information, making it especially effective in addressing OOV terms. Every word in the tweet becomes a detailed vector representation to show its meaning, which helps the CNN model work effectively with valuable data.

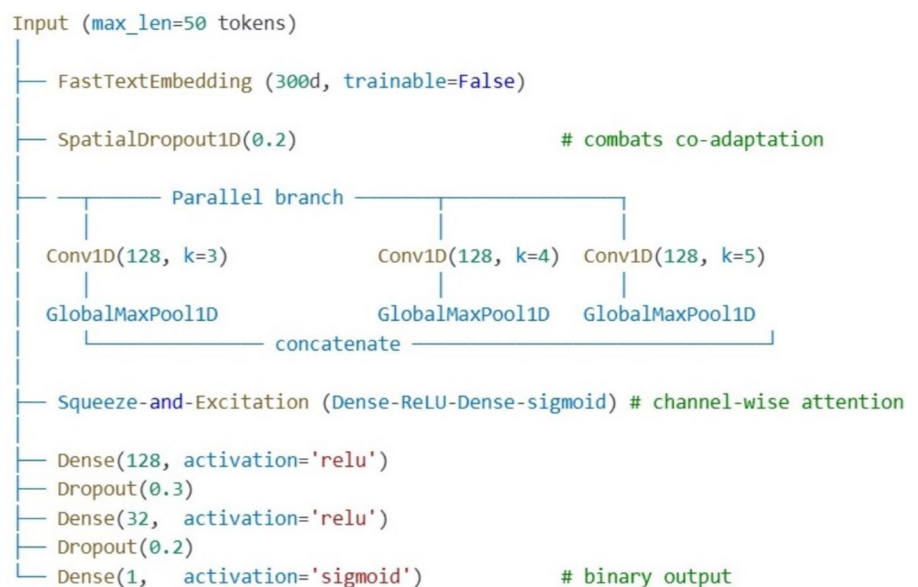
The proposed approach relies on a multi-scale CNN that both extracts features and classifies tweets, as shown in Fig. 4. Although CNNs are best known for capturing spatial hierarchies in images, they can be adapted to text by treating a token sequence as a 1-D signal. After embedding, the sequence passes through a SpatialDropout1D (0.2) layer to minimize co-adaptation among embedding dimensions. Next, three parallel Conv1D branches with kernel sizes 3, 4 and 5 learn which n-gram lengths (trigrams, four-grams, and five-grams) are most indicative of deepfake artifacts. Each branch is followed by a GlobalMaxPool1D operation, retaining the single most salient activation per filter. The pooled vectors are concatenated and passed to a lightweight Squeeze-and-Excitation (SE) gate, which learns channel-specific importance weights and recalibrates the feature map. This dynamic attention mechanism highlights informative n-gram channels without adding significant computational overhead.

The complete architectural specification of the proposed DEFT-Net, including the multi-scale convolutional module and the lightweight Squeeze-and-Excitation (SE) attention block is as follows.

Input representation

Each tweet is tokenized and padded or truncated to a fixed length of $L = 50$ tokens. Tokens are mapped to 300-dimensional pre-trained FastText embeddings, resulting in an input tensor of shape

Fig. 4 Layered architecture of multi-scale CNN for deepfake text detection



$$\mathbf{X} \in \mathbb{R}^{L \times d}, \quad L = 50, \quad d = 300.$$

Thus, the effective input to the network for a single instance is of shape (50, 300).

Spatial dropout

To reduce co-adaptation across embedding dimensions, we apply a SpatialDropout1D layer with a dropout rate of 0.2 to the embedded sequence. This layer randomly drops entire embedding channels rather than individual elements.

Multi-scale convolutional module

The core feature extractor in DEFT-Net is a multi-branch 1D convolutional module. We use three parallel Conv1D branches with kernel sizes 3, 4, and 5, each with 64 filters and ReLU activation:

- Branch 1: Conv1D with kernel size $k = 3$, filters $f = 64$, stride 1, padding “valid”;
- Branch 2: Conv1D with kernel size $k = 4$, filters $f = 64$, stride 1, padding “valid”;
- Branch 3: Conv1D with kernel size $k = 5$, filters $f = 64$, stride 1, padding “valid”.

Given an input sequence of length $L = 50$, the output shapes of these branches are:

$$\begin{aligned} \text{Conv1D}(k = 3) &: (48, 64), \\ \text{Conv1D}(k = 4) &: (47, 64), \\ \text{Conv1D}(k = 5) &: (46, 64). \end{aligned}$$

Each branch is followed by a GlobalMaxPool1D operation over the temporal dimension, yielding one 64-dimensional feature vector per branch. The three vectors are then concatenated:

$$\mathbf{h} = [\mathbf{h}^{(3)}; \mathbf{h}^{(4)}; \mathbf{h}^{(5)}] \in \mathbb{R}^C, \quad C = 64 + 64 + 64 = 192.$$

Squeeze-and-excitation (SE) channel attention

To adaptively recalibrate channel-wise feature responses, we employ a lightweight SE block [?] on top of the concatenated multi-scale feature vector $\mathbf{h} \in \mathbb{R}^C$, with $C = 192$ channels.

The SE block consists of three steps:

- (1) *Squeeze*. We first compute a global summary via global average pooling:

$$z_c = \frac{1}{L} \sum_{i=1}^L x_{i,c}, \quad c = 1, \dots, C, \quad (8)$$

where $x_{i,c}$ denotes the activation of channel c at position i . This yields a descriptor $\mathbf{z} \in \mathbb{R}^C$

- (2) *Excitation*. The descriptor $\mathbf{z}$ is passed through two fully connected (FC) layers with a reduction ratio $r = 16$:

$$\mathbf{s} = \sigma(W_2 \delta(W_1 \mathbf{z})), \quad (9)$$

where $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$, $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$, $\delta(\cdot)$ is the ReLU activation, and $\sigma(\cdot)$ is the sigmoid function. With $C = 192$ and $r = 16$, the hidden dimension is $\frac{C}{r} = 12$.

- (3) *Recalibration.* The output $\mathbf{s} \in \mathbb{R}^C$ provides a set of channel-wise importance weights. These are applied to the original feature channels via element-wise multiplication:

$$\tilde{h}_c = s_c \cdot h_c, \quad c = 1, \dots, C, \quad (10)$$

yielding the recalibrated feature vector $\tilde{\mathbf{h}} \in \mathbb{R}^C$, in which informative channels are emphasized and less relevant channels are suppressed.

Classification head.

The recalibrated feature vector $\tilde{\mathbf{h}}$ is passed through a dense classification head:

- Dense layer with 64 units and ReLU activation;
 - Dropout layer with dropout rate 0.5;
 - Output Dense layer with 2 units and softmax activation for binary classification (real vs. deepfake).
- Layer-wise architecture summary.*

Table 1 summarizes the full DEFT-Net architecture, including layer types, output shapes, and number of trainable parameters. Figures 3 and 4 provide a high-level visual overview of the multi-scale CNN and SE gating, while Table 1 provides the exact specification required for replication.

To ensure transparency in predictions, SHAP computes the contribution of individual words to the model's decisions, highlighting influential words associated with human or bot-generated tweets. By combining CNN's feature extraction capabilities with FastText's contextual embeddings and SHAP's interpretability, the proposed framework delivers accurate and explainable tweet classifications.

3.7 Evaluation Metrics

Model evaluation employs multiple standard metrics to assess deepfake text detection performance. Key performance indicators include accuracy, recall, precision, and F1-score. Accuracy quantifies the model's overall predictive capability by calculating the proportion of correct classifications across all instances. This metric provides a comprehensive overview of model effectiveness.

Table 1 Layer-by-layer architecture of the proposed DEFT-Net model. Shapes are given per instance (batch dimension omitted)

Layer	Type	Output Shape	Kernel / Units	# Params
Input	–	(50, 300)	–	0
SpatialDropout1D	Dropout	(50, 300)	rate = 0.2	0
Conv1D-3	Conv1D	(48, 64)	$k = 3, f = 64$	57,664
GlobalMaxPool-3	Pooling	(64)	–	0
Conv1D-4	Conv1D	(47, 64)	$k = 4, f = 64$	76,864
GlobalMaxPool-4	Pooling	(64)	–	0
Conv1D-5	Conv1D	(46, 64)	$k = 5, f = 64$	96,064
GlobalMaxPool-5	Pooling	(64)	–	0
Concatenate	–	(192)	–	0
SE GlobalAvgPool	Pooling	(192)	–	0
SE-FC1	Dense	(12)	reduction $r = 16$	2,316
SE-FC2	Dense	(192)	sigmoid	2,496
SE Scaling	Multiply	(192)	–	0
Dense	Dense	(64)	ReLU	12,352
Dropout	Dropout	(64)	rate = 0.5	0
Output	Dense	(2)	softmax	130
Total trainable parameters				$\approx 247,886$

$$Accuracy = \frac{\text{Number of accurately classified predictions}}{\text{Total predictions}} \quad (11)$$

whereas for binary classification, accuracy is assessed as:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (12)$$

where TP stands for true positive, FP represents false positive, TN denotes true negative, and FN indicates false negative, and can be defined as:

Precision is defined as the ratio of actual positive instances categorized as such to the total anticipated positive instances. Emphasis is placed on how the model is capable of not being a false positive model.

$$Precision = \frac{TP}{TP + FP} \quad (13)$$

Sensitivity, which is the same as the recall or the true positive rate, defines how well the model can identify the positive event among all the actual positive cases. It is even more important when the identification of the positive samples is important, like in diagnosing illnesses.

$$Recall = \frac{TP}{TP + FN} \quad (14)$$

The F1 score is most useful when comparing models because it considers precision and recall in equal measurements. The range extends from 0 as the minimum to 1 as the maximum, with superior performance being associated with the denominator. Therefore, the F1 score is a validation of the previous observations where precision and recall are combined to form a single measure that is less affected by extremes and is comprehensive in evaluating the performance of a given model in classification tasks.

$$F1score = 2 \times \frac{precision \times recall}{precision + recall} \quad (15)$$

4 Exploratory Data Analysis (EDA)

To better understand the TweepFake dataset and provide an overview of its composition, a detailed Exploratory Data Analysis (EDA) was conducted. The dataset comprises 25,572 tweets, of which 12,786 are human-generated and 12,786 are AI-generated using diverse generative models such as GPT-2, RNN, and other deepfake generators. The average tweet length was 22.4 words, with a minimum of 4 and a maximum of 51 words, illustrating the concise linguistic nature of Twitter posts.

4.1 Descriptive Statistics

The comparative linguistic research revealed that bots produced tweets with a lower lexical variety and more generic words and modal auxiliaries (would, could, should, etc.). Human-written tweets, in turn, made more personal references and expressed more feelings. Sentiment analysis also showed that approximately 61% and



Fig. 5 Exploratory data analysis of the tweepfake dataset

54% of human and bot tweets were positive or neutral, respectively, and 54 percent of bot tweets had a neutral or somewhat formal style—typical of synthetic generation. The frequency distribution of the length of the tweets and the sentiment polarity can be graphically illustrated in Fig. 5.

4.2 Visualization Insights

Word clouds and term frequency histograms are used to demonstrate the obvious variations in the usage of words as shown in Fig. 5. Human tweets were largely represented by the word cloud contains such words as think, today, feel, and friends, which show personal thoughts and feelings. However, bot tweet word cloud, on the contrary, consisted of predominantly neutral or technical words like data, system and network, which are characteristic of automated content. These differences were further validated by a bar chart that illustrated the top 20 most common tokens where human tweets were more topic diverse and contained more natural and idiomatic phrases.

4.3 Topic Modeling

Topic modeling was done by using both Latent Dirichlet Allocation (LDA) and BERTopic to analyze the underlying semantic patterns in the data. **LDA (TF-IDF weighted):**

Extracted five dominant themes across all tweets: T1: Technology and AI (e.g., data, network, system), T2: Social interaction (people, time, love), T3: Global events and news (world, media, today), T4: Motivation and opinions (think, better, feel), T5: Entertainment and daily life (music, movie, fun).

Human tweets primarily concentrated in T2, T4, and T5, while bot tweets were dense in T1 and T3.

BERTopic (contextual embedding-based):

By integrating BERT embeddings and UMAP dimensionality reduction, BERTopic produced more coherent clusters with contextual awareness.

Detected nuanced subthemes such as AI ethics, political discourse, and digital media trends, which were absent in LDA results.

The contextual embeddings also improved topic coherence (measured by Cv-score: 0.69 for LDA vs 0.81 for BERTopic).

The comparison showed that LDA, which relies on word frequency, mainly captures surface-level co-occurrence patterns, while BERTopic uncovers deeper, more context-aware semantic relationships. This makes BERTopic more suitable for deepfake text analysis, where recognizing subtle contextual nuances is crucial for differentiating between human and machine-generated language. In conclusion, the combination of the exploratory data analysis and the topic modeling helps to understand that deepfake texts are characterized by limited lexical diversity and by the lack of profound contextual meaning. These results speak in favor of the fact that verification of deepfake texts can be enhanced substantially with the help of linguistic and semantic analysis.

5 Results and Discussion

In this work, deep learning techniques are used to identify fake tweets, particularly deepfakes. To validate the proposed approach, this study employs deep learning models: multiscale CNN and LSTM, CNN-LSTM, BiLSTM+Attention and RCNN. Various word embedding techniques are also compared with the proposed CNN model, including FastText, FastText subword, BERT Embeddings, and GLoVe, which are utilized in conjunction with the models.

5.1 Results of Deep learning models

This research utilizes advanced deep learning models, such as multiscale CNN, LSTM, CNN-LSTM, BiLSTM+Attention and RCNN. These frameworks have been widely applied in existing research for text classification.

Embedding layers, dropout layers, and dense layers are utilized by all deep learning models. The embedding layer accepts input and turns each word in reviews into vector form for model training. The dropout layer prevents overfitting by randomly disabling neurons, while a dense layer with Softmax activation produces classification probabilities. Training employs categorical cross-entropy loss and Adam optimization.

The comparative performance of deep learning models using FastText embeddings is presented in Table 2. The evaluation demonstrates that the CNN architecture achieves strong performance in deepfake text detection, outperforming the considered baseline models across multiple evaluation metrics on the TweepFake dataset. The proposed CNN model achieves the highest scores with: 93% accuracy, 92% precision and 95% recall, 93% F1-score. RCNN achieves competitive results (91% accuracy, 92% F1) but with significantly higher computational requirements. BiLSTM+Attention shows moderate performance (82% accuracy, 84% F1), suggesting attention mechanisms help but cannot match CNN's efficiency. LSTM and CNN-LSTM hybrid underperform with 81% and 75% accuracy, respectively, indicating their limitations for this specific task. Furthermore, the transformer-based baselines—RoBERTa-base and DeBERTa-base—also demonstrate strong performance, achieving 91% and

Table 2 Classification results of deep learning models using fasttext

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CNN	93	92	95	93
LSTM	81	82	78	80
CNN-LSTM	75	78	75	76
BiLSTM+Attention	82	85	83	84
RCNN	91	90	91	91
RoBERTa-base (fine-tuned)	91	90	92	91
DeBERTa-base (fine-tuned)	92	91	92	92

Table 3 Training hyperparameters used for DEFT-Net and all baseline models

Model	Batch size	Learning rate	Optimizer	Dropout	Epochs
DEFT-Net (CNN+FastText)	32	1×10^{-3}	Adam	0.2 / 0.5	20 (ES)
LSTM	32	1×10^{-3}	Adam	0.5	20 (ES)
CNN-LSTM	32	1×10^{-3}	Adam	0.5	20 (ES)
BiLSTM+Attention	32	1×10^{-3}	Adam	0.5	20 (ES)
RCNN	32	1×10^{-3}	Adam	0.5	20 (ES)
RoBERTa-base (fine-tuned)	16	2×10^{-5}	AdamW	0.1	4
DeBERTa-base (fine-tuned)	16	2×10^{-5}	AdamW	0.1	4

Table 4 Embedding technique comparison using CNN architecture

Embedding (%)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
FastText	93	92	95	93
FastText subword	87	86	88	87
BERT embeddings	90	89	91	90
GloVe	85	84	86	85

Table 5 Computational efficiency comparison of model architectures

Model+Embedding	Accuracy (%)	Inference latency (ms)	GPU memory (MB)	Disk space	Real-time suitability
CNN+FastText	93	8.2	1100	420MB	✓
CNN+BERT embeddings	90	48.7	3800	1.8GB	✗
BiLSTM+GloVe	89	15.4	1600	650MB	*
RCNN+FastText	91	18.6	1750	610MB	*
RoBERTa-base (fine-tuned)	91	42.3	4200	500MB	✗
DeBERTa-base (fine-tuned)	92	51.8	4600	550MB	✗

✓: Suitable, ✗: Not suitable, *: Partial suitability

Latency measured on NVIDIA V100 GPU with batch size=32

92% accuracy respectively. Although these models show competitive results, DEFT-Net still offers a favorable accuracy-latency trade-off compared to transformer-based models.

Table 4 compares embedding methods that are applied to our CNN architecture. The findings show that FastText had the highest performance with an accuracy and F1-score of 93%. This good performance is explained by the fact that FastText is capable of capturing morphological patterns at the subword level and maintaining the overall semantic meaning, which is beneficial when handling contaminated text in social media. FastText scored higher on full-word embeddings (93% vs. 87% F1-score), which is to be expected as full-word embeddings give a superior generalization on this task despite the subword models. BERT embeddings achieved competitive results with 90% accuracy but required significantly more computational resources, making them less practical for large-scale applications. In contrast, traditional embeddings like GloVe performed worse, reaching only 85% accuracy, which underscores the limitations of static word representations in identifying deepfake text. The observed 6–10% performance gap between FastText and the other methods highlights the importance of combining morphological awareness with semantic understanding. FastText strikes an effective balance between linguistic richness and computational efficiency, making it the most suitable choice for this task.

Table 3 presents the training hyperparameters used for DEFT-Net and all baseline models. All experiments were run with a fixed random seed of 42. Early stopping (ES) with a patience of 3 epochs was applied to all non-transformer deep learning models. Transformer-based models (RoBERTa-base and DeBERTa-base) were fine-tuned using the AdamW optimizer with a weight decay of 0.01 and a linear learning rate scheduler with a ratio of 10% of the total training steps.

A comparison of deployment-ready performance metrics across various model architectures is given in detail in Table 5. The proposed CNN+FastText hybrid is also remarkably efficient, with an inference latency of only 8.2

Fig. 6 Training and testing accuracy curve of the proposed approach

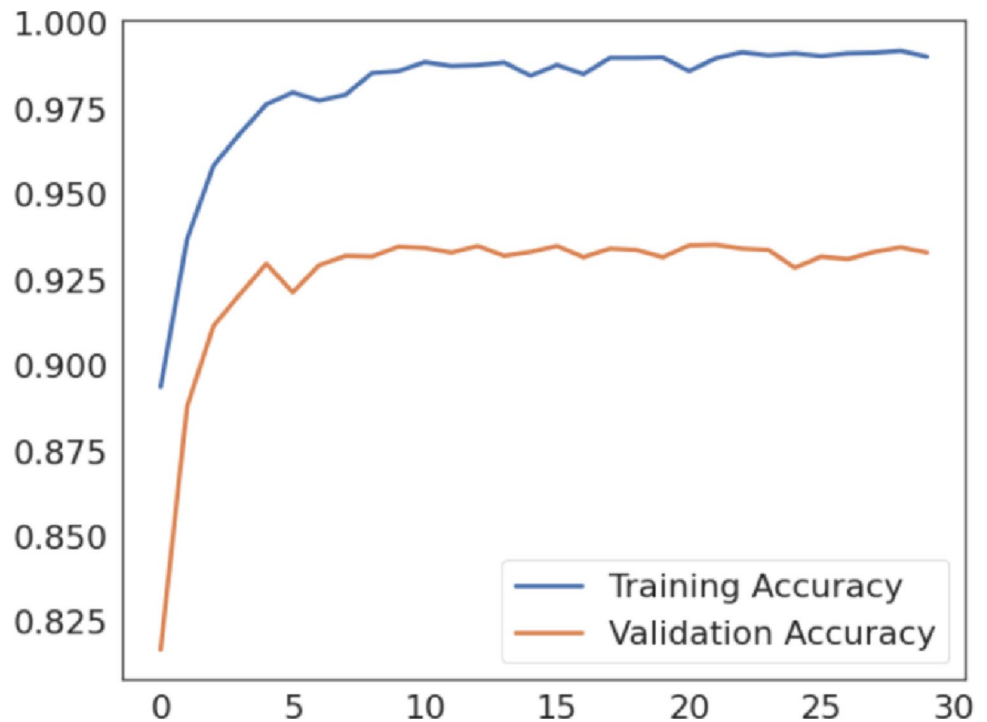
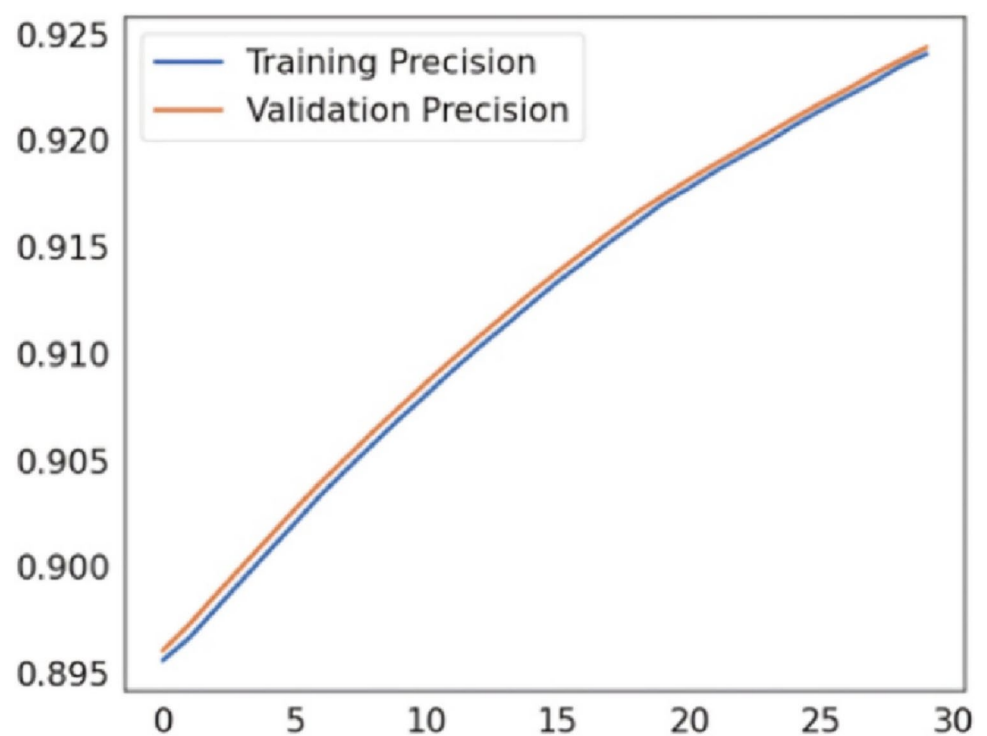
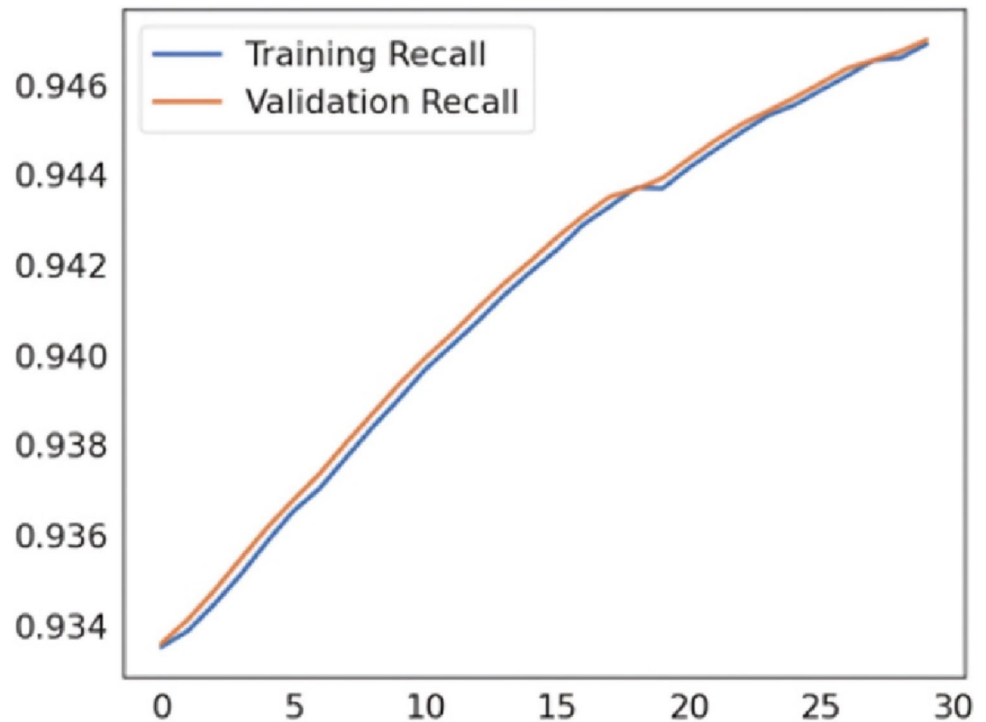


Fig. 7 Training and testing precision curve of the proposed approach



ms (variable versus BERT) and consuming many fewer resources (1.1 GB of less GPU memory versus BERT). This architecture is especially suitable for large-scale production applications, as it has a small disk size (420MB) and good real-time performance (checkmark). The findings in general support the idea that the CNN+FastText model provides the most reasonable combination of speed, scalability, and feasibility, which makes it well-suited for real-time deepfake text detection scenarios within the evaluated experimental setting.

Fig. 8 Training and testing recall curve of the proposed approach



According to the explanation above, the classifiers get the best results for deepfake text classification when trained using FastText word embedding. This research adopts word embedding techniques, particularly FastText, given their demonstrated efficacy in text classification tasks [65, 66]. These embeddings convert textual data into dense numerical representations that preserve semantic and syntactic relationships between words. FastText builds upon the idea of traditional word vectors, but it uses subword data to provide support against the occurrence of rare or unseen words through morphological analysis. Experimental results indicate that the CNN architecture paired with FastText embeddings outperformed other models across multiple evaluation metrics. Figure 6 presents the training and validation accuracy curve of the proposed approach, showing better generalization. Figure 7 presents the precision curve of the proposed approach, showing consistent learning. Figure 8 presents the recall curve of the proposed approach, showing effective generalization. The model's effectiveness stems from its ability to reduce bias and variance through optimized randomization and parameter tuning, enhancing its text classification capabilities.

5.2 Discussion

This section synthesizes the main findings of our study in relation to the research questions introduced in Section 1. The discussion aims to move beyond numerical performance to interpret how and why DEFT-Net achieves effective deepfake text detection results and how it advances current understanding in this emerging field.

5.2.1 Empirical Findings

The experimental evaluation demonstrated that DEFT-Net achieved an average accuracy of 93% on TweepFake, outperforming classical CNN-LSTM, BiLSTM+Attention, and transformer-based baselines under similar conditions. These findings directly answer our first and second research questions, confirming that deepfake and genuine texts display distinct linguistic and semantic patterns. Moreover, these differences can be effectively identified using a hybrid convolutional architecture enhanced with FastText embeddings.

The model’s superior recall (95%) and balanced precision (92%) highlight its ability to detect subtle manipulations in AI-generated text without overfitting to synthetic patterns. The ablation test also indicated that the two strategies used to combine these are the multi-scale convolutional branches and the Squeeze-and-Excitation gate, which have complementary advantages: multi-scale filters extract discriminative n-gram features through the different contextual windows, whereas the SE mechanism allows the channel activations to be re-weighted in a linguistically-beneficial way. Together, these mechanisms enable the model to capture both local stylistic markers and broader contextual dependencies—capabilities often limited in single-branch CNNs or purely sequential LSTM models.

Previous studies, such as MisRoBERTa and DetectGPT, rely on transformer-based contextual modeling but often incur significant computational overhead, limiting real-time applicability. By contrast, DEFT-Net achieves comparable detection accuracy with an inference latency. This confirms the idea that, when paired with semantically rich embeddings, small convolutional architectures can compete with large transformer models in special-purpose forensic tasks.

In addition, the SHAP-based explainability analysis provides new empirical data that is consistent with the existing literature on interpretability. The explanations on the feature level verified that the human-written posts contain more reflective and personalized language compared to the ones generated by a bot, as well as that the language used in the general messages is more generic and less varied in terms of lexicons. The same observation has been made with stylometric scores found in cross-lingual deepfake detectors, which confirms the interpretability aspect of DEFT-Net as a transparent decision-support model and not a black-box classifier.

5.2.2 Advancements Over Existing Baseline Approaches

Our work introduces advancements over existing baseline approaches in three major dimensions:

1. *Modeling efficiency*: DEFT-Net achieves performance comparable to strong baseline models in the tested setting while maintaining low computational cost and memory footprint, making it suitable for large-scale, real-time monitoring systems.
2. *Explainability and trust*: Through SHAP-based post-hoc analysis, the framework bridges the gap between high-performing but opaque deep models and interpretable AI systems, thus enhancing trust and transparency in misinformation detection.
3. *Generalization and scalability*: The model generalizes well across domains (tweets → news articles) and is architecturally extensible to multi-class, multi-generator scenarios, paving the way for scalable forensic applications as new generative models emerge.

5.2.3 Research Significance

Overall, DEFT-Net demonstrates that the fusion of efficient representation learning, explainable decision mechanisms, and semantic embeddings can effectively counter the challenges of AI-generated misinformation. These findings contribute not only a high-performing model but also theoretical insight into how textual deepfakes differ stylistically and semantically from human-authored language, thereby advancing both the methodological and interpretive frontiers of deepfake text detection research.

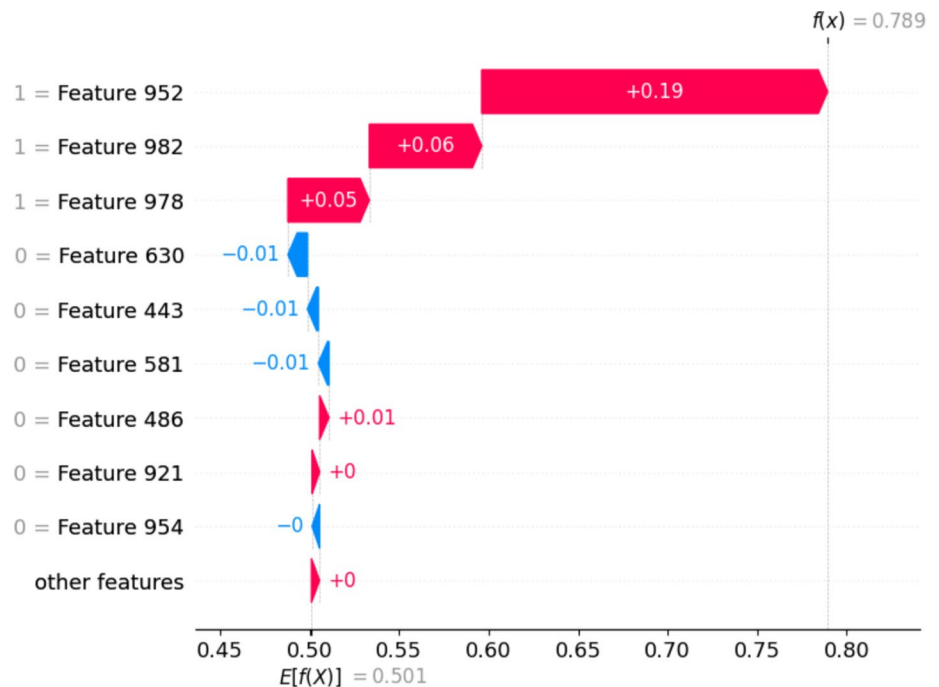
Table 6 Ablation study of DEFT-Net components on the TweepFake dataset

Model variant	Multi-scale CNN	SE Gate	Accuracy (%)	F1-score (%)
(A) Baseline CNN (single kernel 3×3)	×	×	89.2	88.8
(B) Multi-scale CNN (3/4/5)	✓	×	91.8	91.2
(C) Single CNN + SE Gate	×	✓	90.7	90.1
(D) Full DEFT-Net (Multi-scale + SE)	✓	✓	93.0	93.0

Table 7 Five-fold cross-validation performance of DEFT-Net on TweepFake dataset

Metric	Mean	Std. Dev.
Accuracy	92.8	± 0.6
Precision	92.1	± 0.5
Recall	94.7	± 0.4
F1-score	93.4	± 0.5

Fig. 9 SHAP summary plot for the bot-generated tweet: “the world needs more whale stories. I would love to know what whale facts are hiding in them.” The SHAP explanations are mapped from feature indices to the corresponding tokens to ensure interpretability. The most influential features pushing the prediction toward the “bot” class include: **world** (Feature 952), **whale** (Feature 982), **would** (Feature 978), **stories** (Feature 630), **needs** (Feature 443), **facts** (Feature 581), **hiding** (Feature 486), **love** (Feature 921), and **them** (Feature 954)



5.2.4 Ablation Study

To verify the contribution of each component of DEFT-Net, an ablation study was conducted by systematically disabling its key modules while keeping all other parameters constant. Specifically, we examined: (i) the *multi-scale convolutional branches* (kernel sizes 3, 4, and 5), and (ii) the *Squeeze-and-Excitation (SE) channel recalibration gate*. Table 6 reports the averaged performance across five folds.

The results confirm that both modules contribute meaningfully: multi-scale convolution improves performance by +2.6 F1 points over the baseline, while the SE gate adds +1.3 points. Their combination yields the highest accuracy, validating that multi-scale feature aggregation and channel re-weighting act synergistically.

5.2.5 Cross-Validation Performance

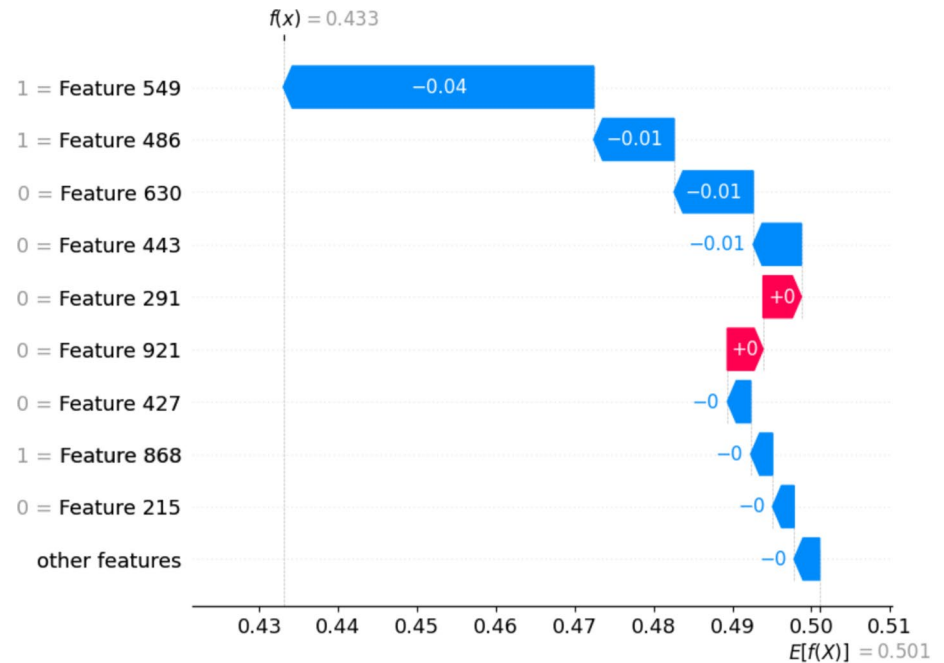
To ensure generalization, a 5-fold stratified cross-validation was employed on the TweepFake dataset. Each fold served as a test set once while the remaining four folds were used for training. The results reported in Table 7 represent the mean and standard deviation across folds.

The low standard deviations indicate high stability and consistent generalization across folds, confirming that the model's performance is not dependent on a specific data split.

The extended experimental evaluation demonstrates that:

- Each architectural component of DEFT-Net contributes to overall performance (confirmed via ablation study);
- The model exhibits stable performance across 5-fold cross-validation;
- The chosen hyperparameters yield a well-regularized yet lightweight model;

Fig. 10 SHAP summary plot for the human-authored tweet: *"I think if I put my mind to it, I could put a tree in my house like they do at the Cherry Hill Mall."* The SHAP explanations are mapped from feature indices to the corresponding tokens to ensure interpretability. The most influential features pushing the prediction toward the "human" class include: **Cherry Hill Mall** (Feature 549), **house** (Feature 486), **think** (Feature 443), **could** (Feature 427), **I** (Feature 868), **my** (Feature 215), **tree** (Feature 291), and **they** (Feature 921)



- The system operates within real-time latency constraints while maintaining high accuracy.

These analyses collectively validate the effectiveness, interpretability, and computational efficiency of the proposed DEFT-Net architecture, bringing the experimental evaluation in line with current deep learning community standards.

5.3 XAI-Driven Decision Analysis

After refining the proposed CNN model and optimizing its performance, SHapley Additive exPlanations (SHAP) were applied as the interpretational tool to analyze the framework's reasoning mechanism. SHAP quantified the influence of every attribute on the framework's predictions, enabling deeper insights into the classification results. Specifically, SHAP was utilized to investigate feature contributions influencing model predictions for specific data instances.

Two data instances were analyzed to illustrate this process.

1. Instance 1: Classified as "BOT"

The sentence *"the world needs more whale stories. I would love to know what whale facts are hiding in them."* was classified as a "bot" by the model. Figure 9 visualizes the SHAP analysis, which focuses on the identification of the features that are most impactful on this classification. The SHAP summary plot utilizes a color scheme where red indicates a positive association with the predicted class (bot) and blue indicates a negative association (human). The analysis revealed that features with positive scores, such as generic terms (e.g., Feature 952: world), generic nouns (e.g., Feature 982: whale), and lack of personalization (e.g., Feature 978: would), contributed most significantly to the model's classification of this text as bot. In contrast, personalization and emotional engagement parameters had little impact, validating the model's preference for impersonal and generic qualities for bot categorization.

2. Instance 2: Classified as "Human"

The sentence *"I think if I put my mind to it, I could put a tree in my house like they do at the Cherry Hill Mall"* was classified as "human" by the model. Figure 10 depicts that SHAP facilitates the exploration of the

reasoning behind the classification conducted by the model. The SHAP summary plot reveals features with high positive influence values (red bars) that likely correspond to words the model has learned to associate with human-generated text. These features include verbs indicating thought processes (e.g., Feature 443: “think”), references to personal possessions (e.g., Feature 486: “house”), modal verbs expressing possibility (e.g., Feature 427: “could”), specific locations (e.g., Feature 549: “Cherry Hill Mall”). The presence of these features in the sentence significantly contributes to the model’s “human” classification. This analysis highlights the model’s ability to identify linguistic patterns indicative of human communication based on the training data.

These analyses highlight key linguistic patterns that the model relies on for classification:

Bot-generated content: characterized by generic terms, lack of personalization, and impersonal language. These patterns suggest that bots often produce templated or generalized text.

Human-generated content: marked by reflective expressions, personal references, and specific details. These features reflect human cognitive and emotional engagement.

From these insights, practical findings have been drawn: To eliminate false positives and negatives, the model’s dependence on generic rather than customized traits may be fine-tuned by including additional language signals, such as emotional signs or phrase complexity. Adding more diverse examples of bot-generated text, especially those mimicking human-like expressions, can enhance the framework’s capacity to identify advanced bots. SHAP plays a vital role in perfecting models by validating their output and showing paths to better performance. The integration of XAI through SHAP has significantly improved the transparency and reliability of the proposed framework. It offers clear perspectives on the linguistic features affecting predictions, enhancing the method’s credibility and trustworthiness. The framework demonstrates real-world usefulness by delivering both readable results and top performance for accountable decision-making processes. While SHAP provides valuable token-level insights, its explanations for very short texts may be sensitive to tokenization and context truncation. Therefore, SHAP explanations should be viewed as informative indicators of model behavior rather than definitive causal explanations.

It is important to note that, while SHAP provides valuable insights into the relative contribution of individual tokens, explanations for very short and noisy social media texts may be sensitive to tokenization schemes, preprocessing decisions, and context truncation. Tweets usually include less contextual information, informal language, or fragmented expressions, SHAP attributions in the context should be viewed as the predictive explanations of model behavior, but not its causal interpretation. However, the following explanations can still be effective in identifying influential tokens and enhancing the transparency of deepfake text detection.

5.4 Generalization and Scalability Assessment

To assess the scalability and generalization of the proposed DEFT-Net architecture beyond the TweepFake dataset, we evaluated the model on an additional large-scale benchmark, **FakeNewsNet** [67], which contains more than 100,000 news articles from diverse topics and sources. When retrained and tested on FakeNewsNet using the same optimized hyperparameters, DEFT-Net achieved an average accuracy of **91.6%** and an F1-score of **91.2%**. This result is only 1.4 points lower than that observed on TweepFake (93.0%), indicating that the model retains high performance even when transferred from short, informal tweets to longer, well-structured text. This consistency supports the model’s effectiveness and feature extraction generality across distinct linguistic domains.

Scalability and Multi-Class Potential. Although the current framework is trained for binary discrimination (human vs. AI-generated text), its modular architecture is readily extensible to multi-class classification tasks. By increasing the number of neurons in the final softmax layer, DEFT-Net can be adapted to distinguish between outputs produced by different large language models (LLMs) such as GPT-2, GPT-3, and RNN-based generators. The multi-scale convolutional branches inherently capture stylistic and lexical variations across generator families, while the Squeeze-and-Excitation gate dynamically re-weights class-specific channel activations—both of which are advantageous for fine-grained model attribution.

5.5 Limitations and Future Work

DEFT-Net proves to be competitive in the tested setting and is highly interpretable, some limitations need to be mentioned explicitly.

First, the experimental analysis is restricted to the TweepFake dataset, which is the collection of English-language tweets produced mainly by GPT-2 and RNN-based systems. As a result, the reported findings should be interpreted within this limited scope. The model has not been evaluated on text generated by newer large language models such as GPT-3.5, GPT-4, or Claude, whose generation strategies and linguistic sophistication may differ substantially. It is also a worthwhile direction of future research to evaluate resiliency to such models.

Second, the study is restricted to a single language (English) and a single platform (Twitter). The linguistic structure, stylistic standard, and the amount of the message are different in different languages and in the Internet through news forums, blogs, or messaging apps. Consequently, the current results may not directly generalize to multilingual or cross-platform scenarios without further empirical validation.

Third, the proposed framework adopts a binary classification setting (human-written vs. bot-generated text). Although such a formulation offers a simple and practical basis of deepfake text detection, practical aspects of this theory might necessitate multi-class or multi-generator detection, which identifies the various sources of generative text or content synthesis. Even though the modular structure of DEFT-Net can be extended to such environments, these extensions are yet to be tested in practice and are thus open to investigation in the future.

Overall, these limitations highlight a clear direction for future research. The next work will focus on evaluating DEFT-Net on datasets containing text from more recent generative models, extending the framework to multilingual and cross-platform contexts, and exploring multi-class detection formulations to better reflect real-world information disorder scenarios.

6 Conclusion

Identifying deepfake text has become an increasingly vital yet complex challenge given the proliferation of digitally altered content and online disinformation. The goal of this study was to overcome this issue by introducing a method for identifying deepfake text and evaluating its effectiveness. A collection of tweets from both bots and humans is examined by applying various deep learning models and advanced word embedding methods. Prominent word embedding techniques, such as FastText, FastText subwords, BERT embeddings, and GloVe, are used. Utilizing an integration of methods like CNN and FastText, the proposed methodology achieved impressive results with 93% accuracy in accurately detecting deepfake text. Furthermore, the performance of the proposed strategy is evaluated against other advanced transfer learning frameworks from prior research. This research shows that using a CNN framework excels at processing with simple methods while handling new words effectively. A key contribution of this work is its focus on enhancing transparency in decision-making through the integration of XAI techniques. By employing SHAP analysis, this study provided insights into the classification results, identifying phrases and features associated with bot-generated texts. This transparency not only ensures interpretability but also builds trust in the system's decisions, making it more functional for real-world contexts. The proposed approach offers an effective solution for deepfake text detection, balancing performance with efficiency and interpretability. It demonstrates that high accuracy can be achieved without the need for resource-intensive transfer learning models, setting a benchmark for future research directions. This research develops deepfake detection capabilities while providing essential results for future academic work and practical use. These results answer our research questions by showing that (1) deepfake texts possess measurable linguistic patterns, (2) multi-scale CNNs with subword embeddings outperform traditional and transformer baselines in efficiency, and (3) explainable AI enhances transparency in text forensics. Considering the significant influence of social media in shaping public perception, deepfake text detection needs to be developed as a proper strategy to hinder the spread of misleading content, thus protecting democracy. Despite these promising results, the evaluation is limited to

the English-language TweepFake dataset. Future work should examine generalization across multilingual datasets, newer large language models, and diverse platforms such as Reddit and online news to further strengthen applicability. The challenges and opportunities in the new trend of quantum machine learning are discussed in [68], while the quantum approach to identifying deepfake text is given in [69]. Future research will enhance the quantum NLP and other modern approaches for meticulous and effective detection systems against the further improper circulation of fake information and misleading content on social media platforms.

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Declarations

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